F statistics

2024-11-11

Load

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

This script uses admixtools @maierAdmixtoolsInferringDemographic2024 to calculate F3 and F4 statistics, using the full global dataset, comprising of 2744 individuals and 198196 loci.

Remove all populations with only one individual

```
pop_tbl <- table(mad_bor_g1000_sum$population)
to_remove <- which(pop_tbl == 1)
to_remove_pops <- names(to_remove)
mad_bor_g1000_sum <-
  mad_bor_g1000_sum %>% filter(!population %in% to_remove_pops)
```

Group by population

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

Create f2 blocks directory

```
if (dir.exists("../data/intermediate/f2")) {
  f2_dir <- "../data/intermediate/f2"
} else {
  dir.create("../data/intermediate/f2")
  gt_extract_f2(mad_bor_g1000_sum,
    outdir = "../data/intermediate/f2",
    fst = FALSE,
    n_cores = 10
  )
  f2_dir <- "../data/intermediate/f2"
}
```

Read in precomputed f2's

```
f2_blocks <- f2_from_precomp(f2_dir,
  afprod = FALSE
)
```

```
## i Reading precomputed data for 136 populations...
## i Reading f2 data for pair 1 out of 9316...i Reading f2 data for pair 2 out of 9316...i Reading f2 da
```

```
dim(f2_blocks)
```

```
## [1]   136   136 45175
```

```
count_snps(f2_blocks)
```

```
## [1] 197972
```

Test for admixture: F3(Malagasy; Asian, African)

```
## Running C = GWD
```

```
## Running C = ESN
```

```
## Running C = MSL
```

```
## Running C = YRI
```

```
## Running C = LWK
```

```
## Running C = Botswana_Kalanga
```

```
## Running C = Zimbabwe_Remba
```

```
## Running C = CAR_Gbaya

## Running C = CAR_DzangaShangaPeople

## Running C = CAR_Mpiemo

## Running C = CAR_Banda

## Running C = CAR_Nzakara

## Running C = CAR_Zande

## Running C = SouthAfrica_ColouredWellington

## Running C = SouthAfrica_Swazi

## Running C = SouthAfrica_Karretjie

## Running C = SouthAfrica_ColouredColesberg

## Running C = SouthAfrica_Xhosa

## Running C = Namibia_Damara

## Running C = Namibia_Himba

## Running C = Angola_Khwe

## Running C = Botswana_Kgalagadi

## Running C = SouthAfrica_Khomani

## Running C = SouthAfrica_SothoSouthern

## Running C = SouthAfrica_Zulu

## Running C = SouthAfrica_Bhaca

## Running C = SouthAfrica_Hlubi

## Running C = SouthAfrica_Mputhi

## Running C = Mozambique_Nhungwe

## Running C = Mozambique_Tsonga

## Running C = Mozambique_Ronga
```

```
## Running C = Mozambique_TongaMoz

## Running C = Mozambique_Chuwabu

## Running C = Mozambique_Tswa

## Running C = Mozambique_Chopi

## Running C = SouthAfrica_Ndebele

## Running C = Angola_Xun

## Running C = Namibia_Wambo

## Running C = SouthAfrica_SothoNorthern

## Running C = Botswana_TswanaBot

## Running C = Botswana_GuiGhanaKgal

## Running C = Lesotho_SothoSouthern

## Running C = Swaziland_Swazi

## Running C = DRC_Manyanga

## Running C = DRC_Yombe

## Running C = DRC_Ndibu

## Running C = DRC_MongoNkundo

## Running C = DRC_LubaLulua

## Running C = DRC_Ntandu

## Running C = DRC_Shi

## Running C = DRC_Tetela

## Running C = DRC_Yans

## Running C = DRC_Mbala

## Running C = DRC_Nandi

## Running C = DRC_Yaka
```

```
## Running C = DRC_Ngombe

## Running C = DRC_Pende

## Running C = DRC_Songe

## Running C = DRC_Kuba

## Running C = DRC_Lunda

## Running C = SouthAfrica_Tsonga

## Running C = SouthAfrica_Tswana

## Running C = SouthAfrica_Venda

## Running C = Rwanda_Nkore

## Running C = Uganda_Konzo

## Running C = Uganda_Kiga

## Running C = Uganda_Fumbira

## Running C = Uganda_Nkore

## Running C = Zambia_Chewa

## Running C = Zambia_Kaonde

## Running C = Zambia_Lozi

## Running C = Zambia_ShonaZam

## Running C = Zambia_Bemba

## Running C = Zambia_Tumbuka

## Running C = Zambia_TongaZam

## Running C = Tanzania_TanzaniaMixed

## Running C = Zanzibar_Swahili

## Running C = Zambia_Ngoni

## Running C = Zambia_Nsenga
```

```
## Running C = SouthAfrica_Coloured

## Running C = Angola_Vasekele

## Running C = Namibia_TsumkweKung

## Running C = Namibia_Herero

## Running C = Namibia_Nama

## Running C = Angola_Sekele

## Running C = Namibia_Kgalagadi

## Running C = DRC_Luozi

## Running C = Zambia_Tswana

## Running C = Zambia_Kalanga

## Running C = Zambia_Kgalagadi

## Running C = Zambia_Mbukushu

## Running C = Zambia_Mbunda

## Running C = Zambia_Kwangwa

## Running C = Zambia_Luyana

## Running C = Zambia_Nkoya

## Running C = Zambia_Nyengo

## Running C = Zambia_Shanjo

## Running C = Zambia_Kwamashi

## Running C = Zambia_Fwe

## Running C = DRC_Ding

## Running C = DRC_Wonk

## Running C = DRC_Lwer

## Running C = DRC_Nsong
```

```
## Running C = DRC_Mbuun

## Running C = DRC_Nzadi

## Running C = DRC_Hungan

## Running C = DRC_Lele

## Running C = DRC_Mpiin

## Running C = DRC_Ngwi

## Running C = DRC_LubaKasai

## Running C = DRC_Ngong

## Running C = DRC_Sakata

## Running C = DRC_Mpe

## Running C = DRC_NunuBobangi

## Running C = DRC_Mbudza

## Running C = DRC_LubaKatanga

## Running C = DRC_Rega

## Running C = DRC_BanguBangu

## Running C = SouthAfrica_Pedi

## Running C = SouthAfrica_Sotho

## Running C = Botswana_Ghanzi
```

```r
test <- do.call(rbind, results)

colnames(test) <- c("Malagasy", "Asian", "African", "est", "se", "z", "p")

test$Asian <- sub("BOR", "Borneo", test$Asian)
test$Asian <- sub("CHS", "Han, China", test$Asian)
test$Asian <- sub("KHV", "Kinh, Vietnam", test$Asian)
test$Asian <- sub("PJL", "Punjabi, Pakistan", test$Asian)
test$Asian <- sub("BEB", "Bengali, Bengal", test$Asian)

test$African <- sub("ESN", "Esan, Nigeria", test$African)
test$African <- sub("YRI", "Yoruba, Nigeria", test$African)
test$African <- sub("GWD", "Mandinka, The Gambia", test$African)
```

```
test$African <- sub("LWK", "Luhye, Kenya", test$African)
test$African <- sub("MSL", "Mende, Sierra Leone", test$African)

test$Malagasy <- sapply(
  strsplit(test$Malagasy, "_"),
  function(x) paste(x, collapse = " ")
)

print(test)
```

```
## # A tibble: 6,776 x 7
##    Malagasy      Asian             African              est      se      z     p
##    <chr>         <chr>             <chr>              <dbl>   <dbl>  <dbl> <dbl>
##  1 Betsileo      Bengali, Bengal   Mandinka, The ~ -0.00369 5.69e-5  -64.8     0
##  2 Betsileo      Borneo            Mandinka, The ~ -0.0110  5.65e-5 -195.      0
##  3 Betsileo      Han, China        Mandinka, The ~ -0.0106  5.86e-5 -180.      0
##  4 Betsileo      Kinh, Vietnam     Mandinka, The ~ -0.0107  5.68e-5 -189.      0
##  5 Betsileo      Punjabi, Pakistan Mandinka, The ~ -0.00243 6.01e-5  -40.4     0
##  6 Betsileo      Sumba             Mandinka, The ~ -0.00940 5.87e-5 -160.      0
##  7 Betsileo      Timor             Mandinka, The ~ -0.00815 7.21e-5 -113.      0
##  8 Betsimisaraka Bengali, Bengal   Mandinka, The ~ -0.00546 4.92e-5 -111.      0
##  9 Betsimisaraka Borneo            Mandinka, The ~ -0.0104  5.40e-5 -192.      0
## 10 Betsimisaraka Han, China        Mandinka, The ~ -0.0101  5.52e-5 -182.      0
## # i 6,766 more rows
```

Admixture f3 statistics were used in the configuration f3(Malagasy; Asian, African) to examine which comparative African and Asian populations show the greatest similarity to the 'source' populations of the Malagasy.

```
write.table(test,
  file = paste0(res_dir, "f3_results_all_combinations"),
  row.names = FALSE, quote = FALSE, sep = "\t"
)
```

Examine cases where f3 value is not negative

```
test <- read.table(paste0(res_dir, "f3_results_all_combinations"),
  header = TRUE, sep = "\t"
)
fail <- which(test$est > 0)
test_fail <- test[fail, ]

test_fail$poor_fit <- NA

check_refs <- function(test_fail) {
  poor_fit_asian_pop <- c("Punjabi, Pakistan", "Bengali, Bengal")

  # check if this row contains any of the poor fit populations
  if (any(test_fail["Asian"] %in% poor_fit_asian_pop) || any(grepl("SouthAfrica", test_fail["African"]))
    # add "Yes" to the poor_fit column
    test_fail["poor_fit"] <- "Yes"
```

```
  } else {
    # add "No" to the poor_fit column
    test_fail["poor_fit"] <- "No"
  }
}
test_fail$poor_fit <- apply(test_fail, check_refs, MARGIN = 1)
any(test_fail$poor_fit == "No")
```

```
## [1] FALSE
```

All use either Punjabi, Pakistan or Bengali, Bengal as the Asian reference population, or a mixed ancestry South African population as the African reference population, which are likely to be poor fits for the source populations of the Malagasy.

```
failz <- which(test$z > -2)
test_failz <- test[failz, ]
test_failz$poor_fit <- NA
check_refs <- function(test_failz) {
  poor_fit_asian_pop <- c("Punjabi, Pakistan", "Bengali, Bengal")

  # check if this row contains any of the poor fit populations
  if (any(test_failz["Asian"] %in% poor_fit_asian_pop) || any(grepl("SouthAfrica", test_failz["African"]
    # add "Yes" to the poor_fit column
    test_failz["poor_fit"] <- "Yes"
  } else {
    # add "No" to the poor_fit column
    test_failz["poor_fit"] <- "No"
  }
}
test_failz$poor_fit <- apply(test_failz, check_refs, MARGIN = 1)
any(test_failz$poor_fit == "No")
```

```
## [1] FALSE
```

Examine min F3 values for each Malagasy popualtion

```
obj <- test %>%
  group_by(Malagasy) %>%
  slice(which.min(est))
obj
```

```
## # A tibble: 8 x 7
## # Groups:   Malagasy [8]
##   Malagasy           Asian  African              est        se      z     p
##   <chr>              <chr>  <chr>              <dbl>     <dbl>  <dbl> <dbl>
## 1 Betsileo           Borneo Mozambique_Ronga -0.0124 0.0000925 -134.      0
## 2 Betsimisaraka      Borneo Mozambique_Chopi -0.0120 0.0000810 -149.      0
## 3 Diego              Borneo Mozambique_Tswa  -0.0113 0.0000828 -137.      0
## 4 Merina             Borneo Mozambique_Tswa  -0.0112 0.0000747 -149.      0
```

```
## 5 Mikea              Borneo Mozambique_Ronga -0.0105 0.000111   -94.7     0
## 6 Sakalava Tsimihety Borneo Mozambique_Chopi -0.0122 0.0000811 -150.      0
## 7 Temoro             Borneo Mozambique_Ronga -0.0120 0.000104  -115.      0
## 8 Vezo               Borneo Mozambique_Chopi -0.0120 0.0000829 -145.      0
```

```
write.table(
  file = paste0(res_dir, "minimum_f3_malagasy_pops"),
  obj, row.names = FALSE, quote = FALSE, sep = "\t"
)
```

```
# Examine z-scores remembering that z-score is affected by population size
min_z <- test %>%
  group_by(Malagasy) %>%
  slice(which.min(z))
min_z
```

```
## # A tibble: 8 x 7
## # Groups:   Malagasy [8]
##   Malagasy           Asian  African       est        se     z     p
##   <chr>              <chr>  <chr>       <dbl>     <dbl> <dbl> <dbl>
## 1 Betsileo           Borneo DRC_Shi  -0.0115  0.0000551 -209.     0
## 2 Betsimisaraka      Borneo DRC_Shi  -0.0110  0.0000516 -213.     0
## 3 Diego              Borneo DRC_Shi  -0.0102  0.0000485 -211.     0
## 4 Merina             Borneo DRC_Shi  -0.0105  0.0000567 -185.     0
## 5 Mikea              Borneo DRC_Ngwi -0.00991 0.0000614 -161.     0
## 6 Sakalava Tsimihety Borneo DRC_Shi  -0.0112  0.0000508 -220.     0
## 7 Temoro             Borneo DRC_Shi  -0.0110  0.0000537 -205.     0
## 8 Vezo               Borneo DRC_Shi  -0.0110  0.0000546 -201.     0
```

```
mad_bor_g1000_sum %>%
  filter(population == "DRC_Ngwi") %>%
  nrow()
```

```
## [1] 67
```

```
mad_bor_g1000_sum %>%
  filter(population == "DRC_Shi") %>%
  nrow()
```

```
## [1] 129
```

```
mad_bor_g1000_sum %>%
  filter(population == "Mozambique_Ronga") %>%
  nrow()
```

```
## [1] 2
```

```
mad_bor_g1000_sum %>%
  filter(population == "Mozambique_Chopi") %>%
  nrow()
```

```
## [1] 4
```

```
mad_bor_g1000_sum %>%
  filter(population == "Mozambique_Tswa") %>%
  nrow()
```

```
## [1] 4
```

```
write.table(
  file = paste0(res_dir, "minimum_z-score_malagasy_pops"),
  min_z, row.names = FALSE, quote = FALSE, sep = "\t"
)
```

F3 values for each Malagasy population descending all combinations

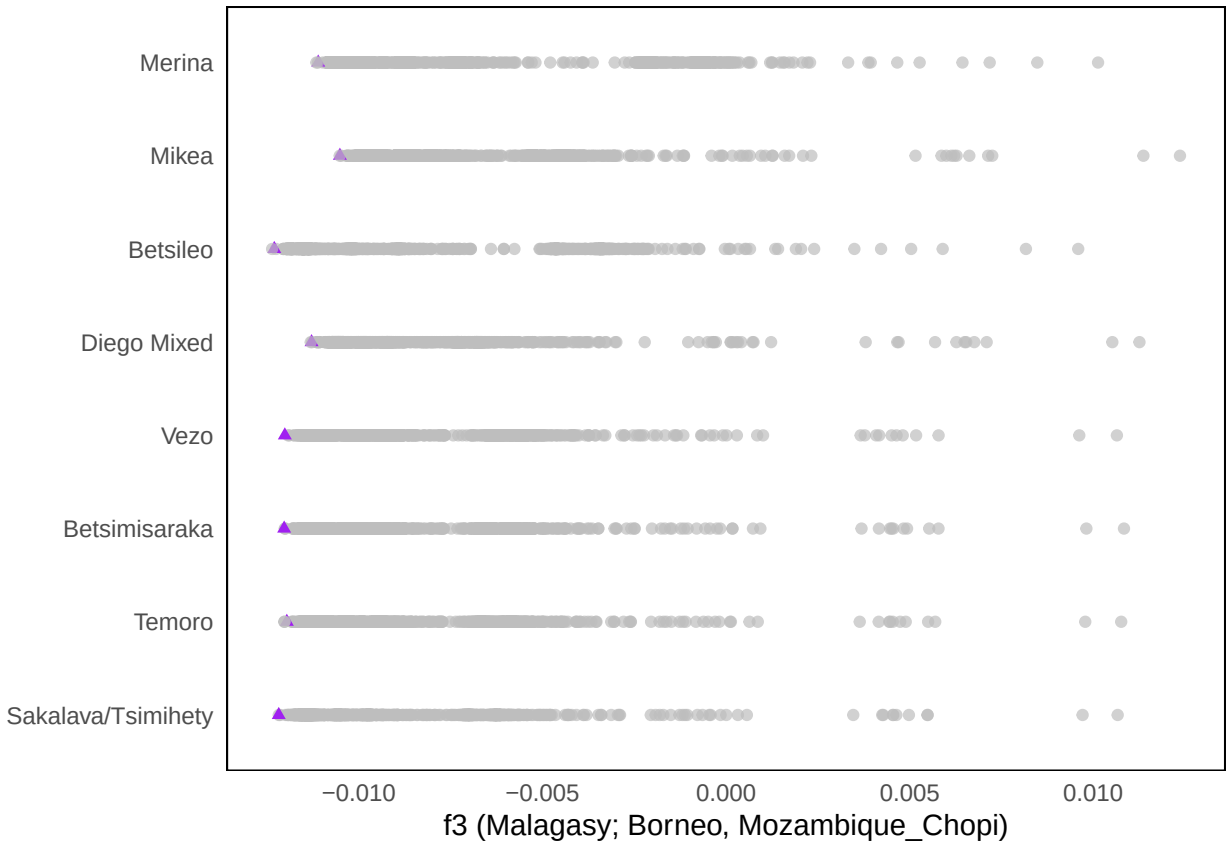
```
# update labels
test <- test %>% mutate(Malagasy = case_when(
  Malagasy == "Sakalava Tsimihety" ~ "Sakalava/Tsimihety",
  TRUE ~ Malagasy
))
test <- test %>% mutate(Malagasy = case_when(
  Malagasy == "Diego" ~ "Diego Mixed",
  TRUE ~ Malagasy
))

test_grouped <- test %>%
  select(est, se, z, Malagasy, Asian, African) %>%
  reframe(est = est, se = se, z, Malagasy, Asian, African)

test_grouped$highlight <- "NA"
# find rows with"Mozambique_Chopi" and"Borneo" as references
test_grouped$highlight[which(test_grouped$African == "Mozambique_Chopi" & test_grouped$Asian == "Borneo
test_grouped$highlight <- as.factor(test_grouped$highlight)

test_grouped %>%
  group_by(Malagasy) %>%
  arrange(desc(est)) %>%
  ggplot(aes(
    x = est, y = reorder(Malagasy, est),
    color = highlight, shape = highlight, alpha = highlight
  )) +
  geom_point() +
  labs(
    x = "f3 (Malagasy; Borneo, Mozambique_Chopi)", y = NULL,
    color = element_blank(), shape = element_blank(),
    alpha = element_blank()
  ) +
  scale_shape_manual(
    values = c(
```

```
    "NA" = 19, # "Borneo" = 3, "Mozambique_Chopi" = 15,
    "Borneo and Mozambique_Chopi" = 17
  ),
  breaks = c(
    # "Borneo", "Mozambique_Chopi",
    "Borneo and Mozambique_Chopi"
  )
) +
scale_color_manual(
  values = c(
    "NA" = "grey", # "Borneo" = "lightpink", "Mozambique_Chopi" = "lightblue",
    "Borneo and Mozambique_Chopi" = "purple"
  ),
  breaks = c(
    # "Borneo", "Mozambique_Chopi",
    "Borneo and Mozambique_Chopi"
  )
) +
scale_alpha_manual(
  values = c(
    "NA" = 0.7, # "Borneo" = 0.7, "Mozambique_Chopi" = 0.7,
    "Borneo and Mozambique_Chopi" = 1
  ),
  breaks = c(
    # "Borneo", "Mozambique_Chopi",
    "Borneo and Mozambique_Chopi"
  )
) +
theme_minimal() +
theme(
  legend.position = "none",
  axis.line = element_blank(),
  panel.grid.major = element_blank(),
  panel.grid.minor = element_blank(),
  panel.background = element_rect(fill = "white")
)
```




```
ggsave(
  filename = paste0(res_dir, "F3_all_combinations.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

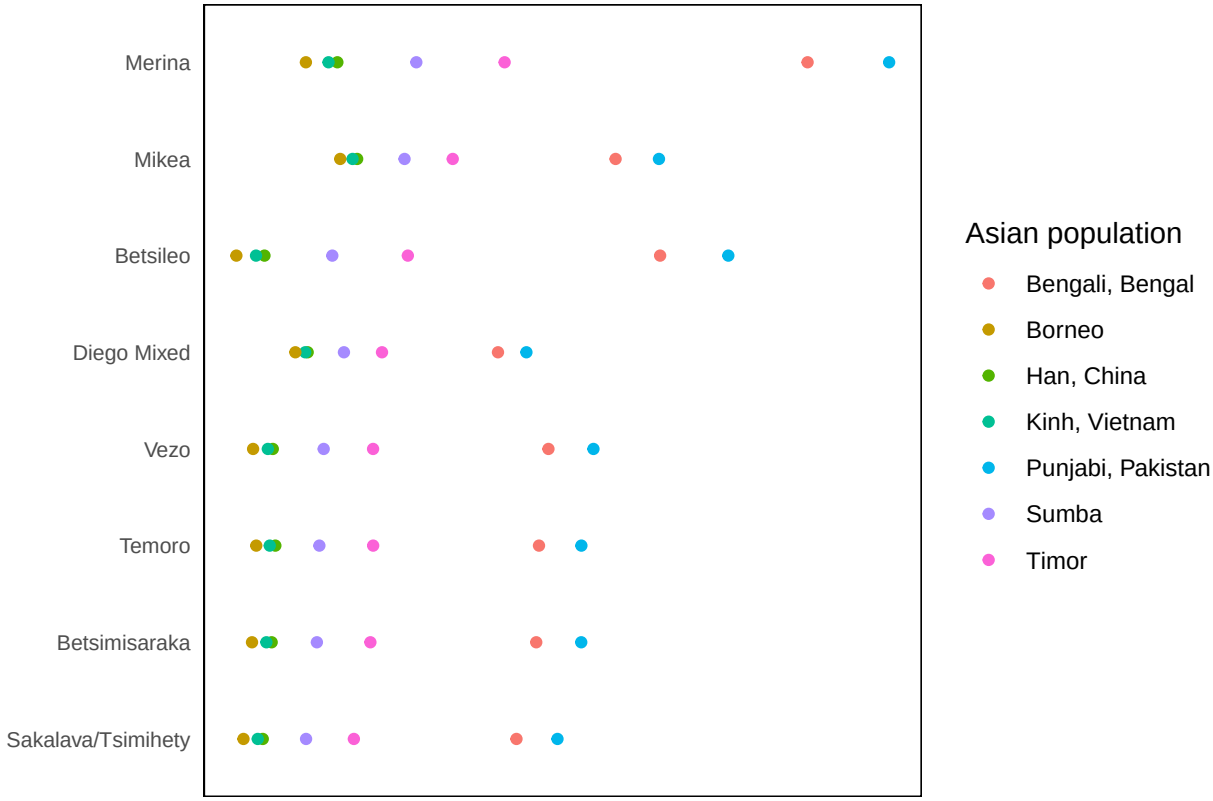
Examining f3 for each Asian reference population

F3 values for each Malagasy population grouped, using Mozambique_Chopi as the African reference.

```
adm_f3 <- test %>%
  filter(African == "Mozambique_Chopi") %>%
  select(est, se, z, Malagasy, Asian) %>%
  reframe(est = est, se = se, z, Malagasy, Asian)


adm_f3 %>%
  group_by(Malagasy) %>%
  arrange(desc(est)) %>%
  ggplot(aes(x = est, y = reorder(Malagasy, est), colour = Asian)) +
  geom_point() +
  labs(
    x = "f3 (Malagasy; Asian reference, Mozambique_Chopi)", y = NULL,
    colour = "Asian population"
  ) +
```

```
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```



f3 (Malagasy; Asian reference, Mozambique_Chopi)

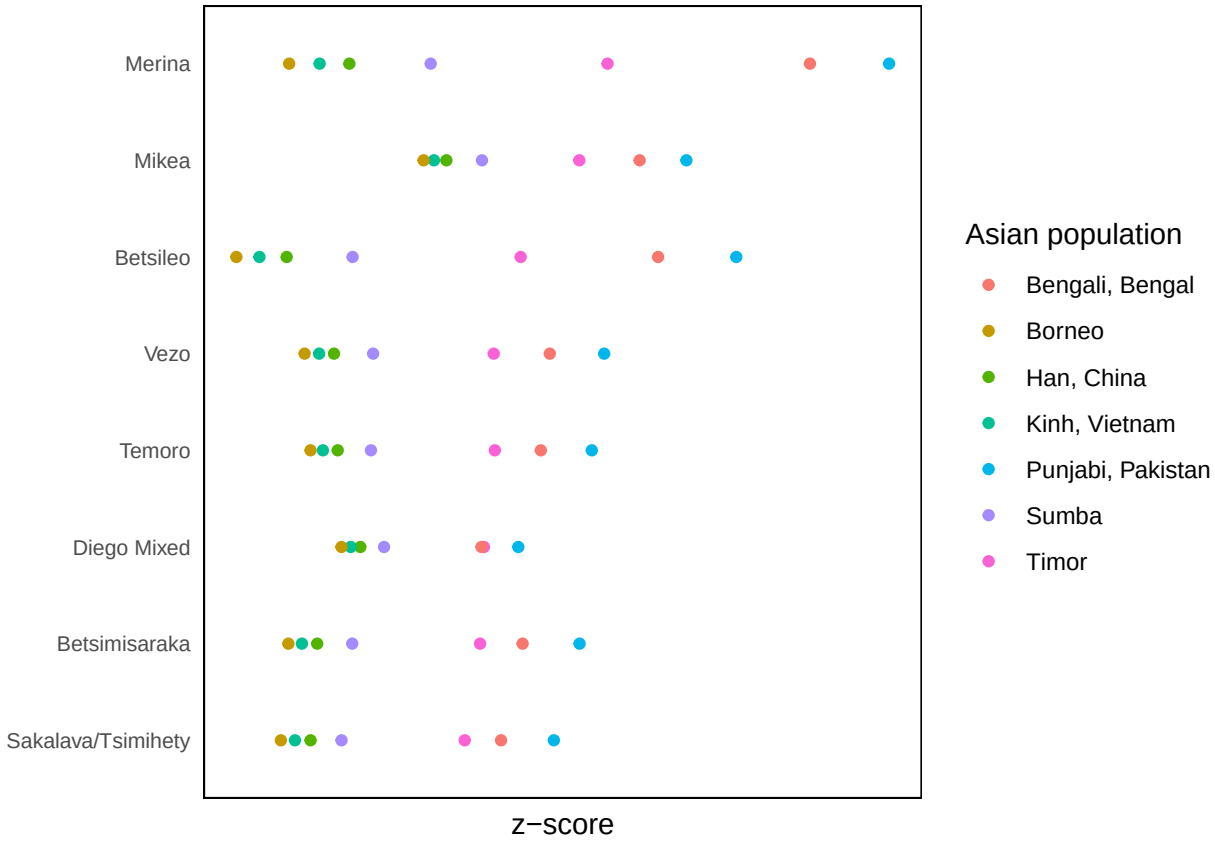
```
ggsave(
  filename = paste0(res_dir, "F3_Asian_source_per_malagasy_pop.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

Also check z-scores below

```
adm_f3 <- test %>%
  filter(African == "Mozambique_Chopi") %>%
  select(est, se, z, Malagasy, Asian) %>%
  reframe(est = est, se = se, z, Malagasy, Asian)


adm_f3 %>%
```

```
  group_by(Malagasy) %>%
  arrange(desc(z)) %>%
  ggplot(aes(x = z, y = reorder(Malagasy, z), colour = Asian)) +
  geom_point() +
  labs(
    x = "z-score", y = NULL,
    colour = "Asian population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```




```
ggsave(
  filename = paste0(res_dir, "F3_Asian_source_per_malagasy_pop_z.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

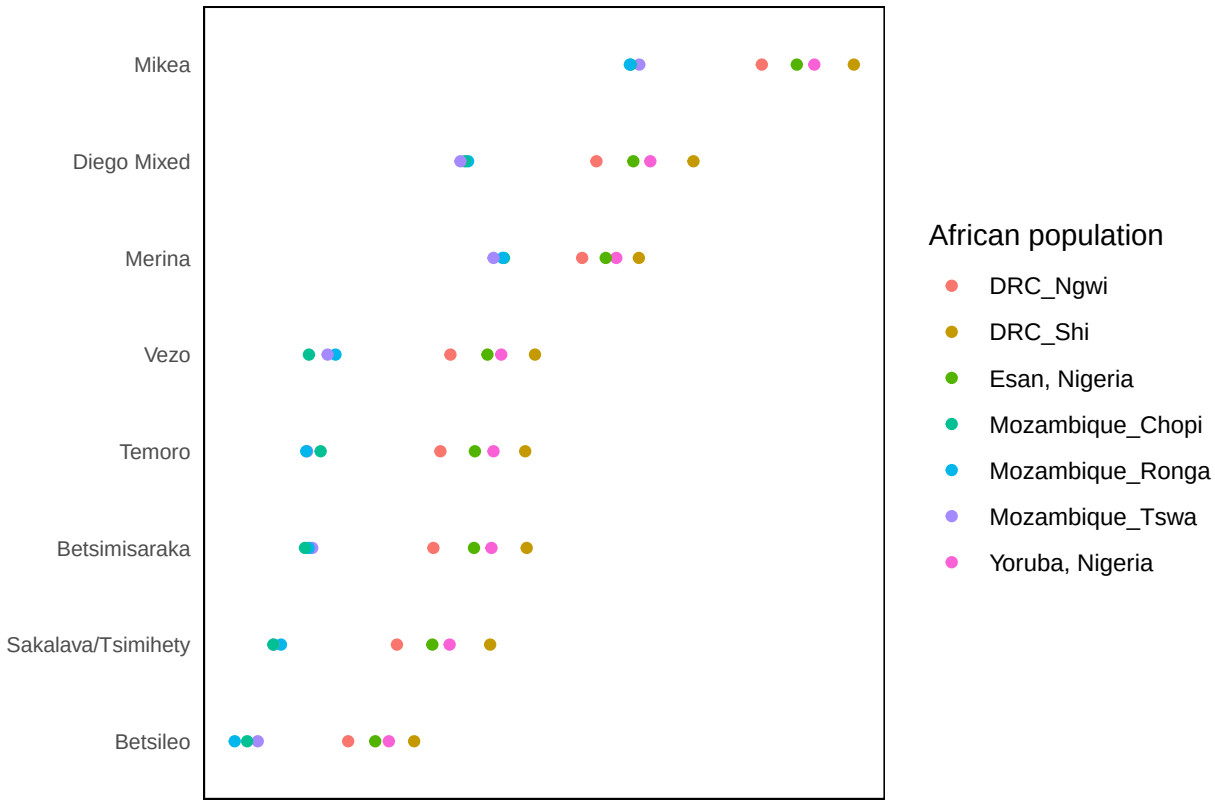
Examining f3 for each African reference population

There are many African popualtions in the dataset, here we will focus on the ones with the lowest f3 values, which are likely to be the most similar to the African source population for the Malagasy.

```
adm_f3 <- test %>%
  filter(Asian == "Borneo") %>%
  filter(African %in% c(
    "Mozambique_Chopi", "Mozambique_Ronga", "Mozambique_Tswa",
    "DRC_Shi", "DRC_Ngwi", "Yoruba, Nigeria", "Esan, Nigeria"
  )) %>%
  reframe(est = est, se = se, z, Malagasy, African)

plt <- adm_f3 %>%
  group_by(Malagasy) %>%
  arrange(desc(est)) %>%
  ggplot(aes(x = est, y = reorder(Malagasy, est), colour = African)) +
  geom_point() +
  labs(
    x = "f3 (Malagasy; Borneo, African reference)", y = NULL,
    colour = "African population"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )

plt
```



f3 (Malagasy; Borneo, African reference)

```
ggsave(
  filename = paste0(res_dir, "F3_African_source_per_malagasy_pop.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

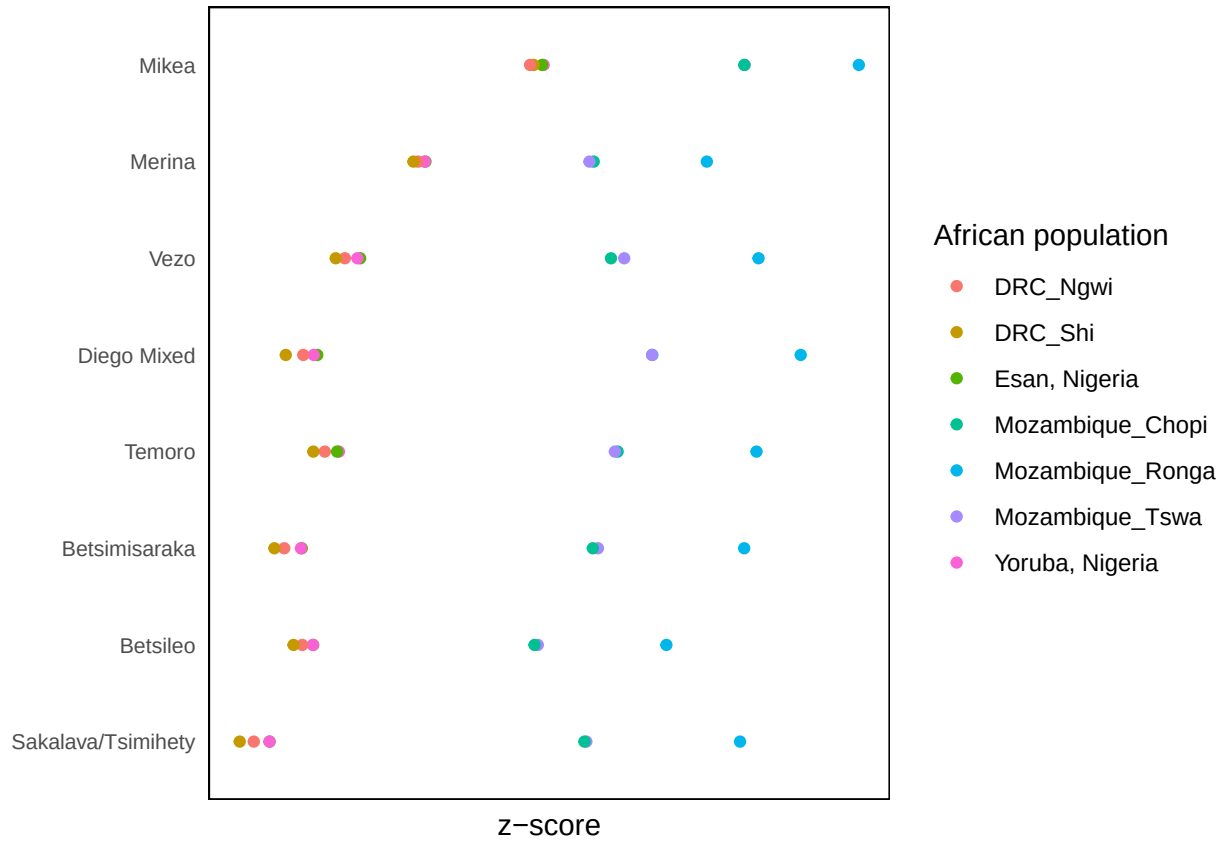
Also check z-scores

```
adm_f3 <- test %>%
  filter(Asian == "Borneo") %>%
  filter(African %in% c(
    "Mozambique_Chopi", "Mozambique_Ronga", "Mozambique_Tswa",
    "DRC_Shi", "DRC_Ngwi", "Yoruba, Nigeria", "Esan, Nigeria"
  )) %>%
  select(est, se, z, Malagasy, African) %>%
  reframe(est = est, se = se, z, Malagasy, African)


plt <- adm_f3 %>%
  group_by(Malagasy) %>%
  arrange(desc(z)) %>%
  ggplot(aes(x = z, y = reorder(Malagasy, z), colour = African)) +
  geom_point() +
  labs(
    x = "z-score", y = NULL,
    colour = "African population"
```

```
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.text.y = element_text(size = 8),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )

plt
```




```
ggsave(
  filename = paste0(res_dir, "F3_African_source_per_malagasy_pop_z.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

Minimum z-scores were observed using the DRC_Shi or DRC_Ngwi populations (n = 129 and n = 67 respectively), likely due to their much larger sample size in comparison to the Mozambique populations (Mozambique_Ronga, n = 2; Mozambique_Chopi, n = 4; Mozambique_Tswa, n = 4).

F4 ratio

To estimate the proportion of each ancestry source within the Malagasy population, we will use an f4 ratio in the following configurations:

f4(GBR,Gorilla;Malagasy,Borneo)/f4(GBR,Chimpanzee;Esan,Borneo)

f4(GBR,Gorilla;Malagasy,Esan)/f4(GBR,Chimpanzee;Borneo,Esan)

Load

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

Split Merina

```
qmats <- read_q_files("../data/intermediate/admixture")

fam <- read.table(
  "../data/intermediate/admixture/admixture_supervised_subset.fam",
  header = FALSE
)

qmats$id <- fam$V2

# subset mad_bor_g1000_sum to only individuals in qmats$id
mad_bor_g1000_sum_sub <- mad_bor_g1000_sum %>%
  filter(id %in% qmats$id)

# tidy with the gt
mad_bor_g1000_sum_sub <- augment(get_q_matrix(qmats, k = 2, run = 1),
  data = mad_bor_g1000_sum_sub
)
# filter to merina
merina <- mad_bor_g1000_sum_sub %>% filter(population == "Merina")
```

```r
# save output
saveRDS(merina, file = paste0(res_dir, "merina_qmats.rds"))

# find which .Q column represents Asian ancestry
asian_ancestry_q <- mad_bor_g1000_sum_sub %>%
  filter(population == "Borneo") %>%
  # Get median value of Q1 and Q2
  summarise(.Q1 = median(.Q1), .Q2 = median(.Q2)) %>%
  # Pivot in order to subset
  pivot_longer(
    cols = c(".Q1", ".Q2"),
    names_to = "column_name",
    values_to = "median_val"
  ) %>%
  # Find the row with the max value
  slice_max(median_val, n = 1) %>%
  # get the column name
  pull(column_name)

# split groups
merina$population <- ifelse(merina %>% select(all_of(asian_ancestry_q)) > 0.5,
  "MerinaA",
  "MerinaB"
)
merina <- merina[, c("id", "population")]

# match V2 and merina$id and replace the population column in merina_fam with
# the population column in merina
matching_m <- match(mad_bor_g1000_sum$id, merina$id)
mad_bor_g1000_sum$population <- ifelse(!is.na(matching_m),
  merina$population[matching_m], mad_bor_g1000_sum$population
)

# count number of MerinaA
cat(
  "Merina group > 0.5",
  length(which(mad_bor_g1000_sum$population == "MerinaA"))
)
```

```
## Merina group > 0.5 23
```

```r
cat(
  "Merina group < 0.5",
  length(which(mad_bor_g1000_sum$population == "MerinaB"))
)
```

```
## Merina group < 0.5 11
```

Remove all populations with only one individual

```
pop_tbl <- table(mad_bor_g1000_sum$population)
to_remove <- which(pop_tbl == 1)
to_remove_pops <- names(to_remove)
mad_bor_g1000_sum <-
  mad_bor_g1000_sum %>% filter(!population %in% to_remove_pops)
```

Group by population

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

Add Chimpanzee outgroup for f4

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% ungroup()
chimp <- gt_load("../data/intermediate/pan_tro_liftover/chimp_ref.gt")
f4_merged_data <- rbind(mad_bor_g1000_sum, chimp,
  as_is = FALSE, flip_strand = TRUE,
  quiet = FALSE, use_position = FALSE,
  backingfile = tempfile()
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ----------------------------
## dataset: reference
## number of SNPs: 198196 reduced to 180431
## ( 0 are ambiguous, of which 0  were removed)
## ----------------------------
## dataset: target
## number of SNPs: 180431 reduced to 180431
## ( 0 were flipped to match the reference set)
## ( 0 are ambiguous, of which 0 were removed)

##
## gen_tibble saved to /tmp/Rtmpdilq8g/filefb5be70798612.gt

## using FBM RDS: /tmp/Rtmpdilq8g/filefb5be70798612.rds

## with FBM backing file: /tmp/Rtmpdilq8g/filefb5be70798612.bk

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/tmp/Rtmpdilq8g/filefb5be70798612.gt')
```

```r
g1000 <- gt_load("../data/raw/data_input/1000_Genomes_b37_clean.gt")
# append population info
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")

pop_table_sub <- subset(pop_table, pop_table$IID %in% g1000$id)

matching_m <- match(g1000$id, pop_table_sub$IID)
g1000$population <- ifelse(!is.na(matching_m),
  pop_table_sub$V2[matching_m],
  g1000$population
)

british <- g1000 %>% filter(population == "GBR")

show_loci(british)$name <- paste(show_loci(british)$chromosome,
  show_loci(british)$position,
  sep = "_"
)

f4_merged_data <- rbind(f4_merged_data, british,
  as_is = FALSE, flip_strand = TRUE,
  quiet = FALSE, use_position = FALSE,
  backingfile = tempfile()
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 180431 reduced to 180431
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 670552 reduced to 180431
## ( 31297 were flipped to match the reference set)
## ( 3 are ambiguous, of which 3 were removed)

##
## gen_tibble saved to /tmp/Rtmpdilq8g/filefb5be31dccf2f.gt

## using FBM RDS: /tmp/Rtmpdilq8g/filefb5be31dccf2f.rds

## with FBM backing file: /tmp/Rtmpdilq8g/filefb5be31dccf2f.bk

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('/tmp/Rtmpdilq8g/filefb5be31dccf2f.gt')
```

```r
nrow(f4_merged_data)
```

```
## [1] 2792
```

```r
f4_merged_data <- f4_merged_data %>% group_by(population)
```

Create f2 blocks directory

```r
if (dir.exists("../data/intermediate/f2_outgroup")) {
  f2_dir <- "../data/intermediate/f2_outgroup"
} else {
  dir.create("../data/intermediate/f2_outgroup")
  gt_extract_f2(f4_merged_data,
    outdir = "../data/intermediate/f2_outgroup"
  )
}
```

Read in precomputed f2's

```r
f2_dir <- "../data/intermediate/f2_outgroup"
f2_blocks <- f2_from_precomp(f2_dir, afprod = TRUE)
```

```
## i Reading precomputed data for 139 populations...
## i Reading ap data for pair 1 out of 9730...i Reading ap data for pair 2 out of 9730...i Reading ap da
```

```r
dim(f2_blocks)
```

```
## [1]   139   139 41399
```

```r
count_snps(f2_blocks)
```

```
## [1] 180229
```

F4 ratio: estimating the proportion of African admixture

Sister taxon and outgroup on the left, target in the centre, and sister population and source of admixture on the right.

```r
malagasy <- c(
  "Betsimisaraka", "MerinaA", "MerinaB", "Diego", "Sakalava_Tsimihety",
  "Betsileo", "Temoro", "Vezo", "Mikea"
)
```

```r
pop_matrix <-
  as.matrix(data.frame("GBR", "chimp_ref", malagasy, "Borneo", "ESN"))

qp4_african_p <- qpf4ratio(f2_blocks, pop_matrix)

qp4_african_p$pop3 <- factor(qp4_african_p$pop3,
  levels = c(
    "Betsimisaraka", "MerinaA", "MerinaB", "Diego",
    "Sakalava_Tsimihety", "Betsileo",
    "Temoro", "Vezo", "Mikea"
  )
)
# order by qp4_african_p$pop3 levels
qp4_african_p <- qp4_african_p %>% mutate(pop3 = forcats::fct_inorder(pop3))
# round alpha to 1 decimal place
qp4_african_p
```

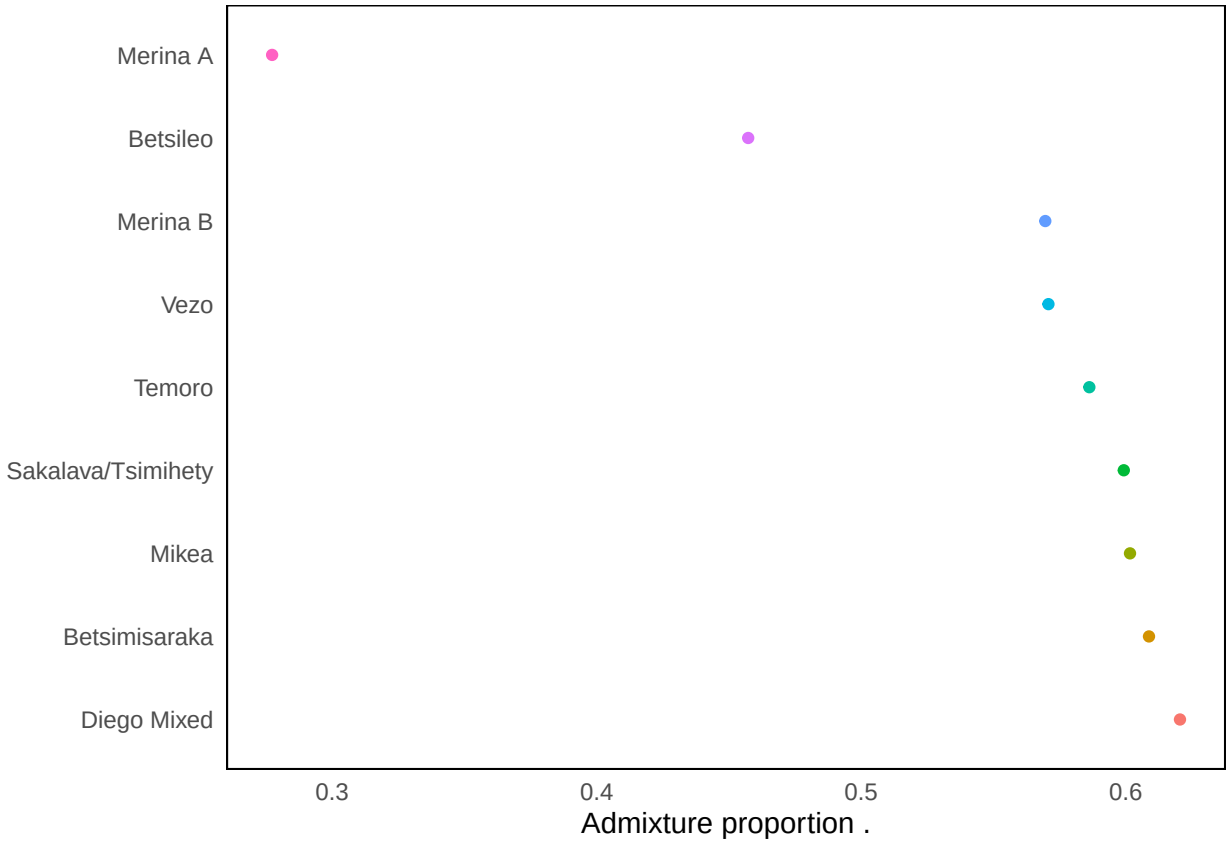
```
## # A tibble: 9 x 8
##   pop1  pop2      pop3               pop4   pop5  alpha      se     z
##   <chr> <chr>     <fct>              <chr>  <chr> <dbl>   <dbl> <dbl>
## 1 GBR   chimp_ref Betsimisaraka      Borneo ESN   0.609 0.00567 107.
## 2 GBR   chimp_ref MerinaA            Borneo ESN   0.277 0.00741  37.4
## 3 GBR   chimp_ref MerinaB            Borneo ESN   0.570 0.00828  68.8
## 4 GBR   chimp_ref Diego              Borneo ESN   0.621 0.00561 111.
## 5 GBR   chimp_ref Sakalava_Tsimihety Borneo ESN   0.599 0.00547 110.
## 6 GBR   chimp_ref Betsileo           Borneo ESN   0.457 0.00594  76.9
## 7 GBR   chimp_ref Temoro             Borneo ESN   0.586 0.00610  96.1
## 8 GBR   chimp_ref Vezo               Borneo ESN   0.571 0.00627  91.0
## 9 GBR   chimp_ref Mikea              Borneo ESN   0.602 0.00717  83.9
```

```r
# save output
write.table(
  file = paste0(res_dir, "f4_ratio_african_admixture"),
  qp4_african_p, row.names = FALSE, quote = FALSE, sep = "\t"
)
```

```r
# update labels
# rename qp4_african_p$pop3 "Sakalava Tsimihety" to "Sakalava/Tsimihety"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "Diego" to "Diego Mixed"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "Diego" ~ "Diego Mixed",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "MerinaA" to "Merina A"
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "MerinaA" ~ "Merina A",
  TRUE ~ pop3
))
# rename qp4_african_p$pop3 "MerinaB" to "Merina B"
```

```r
qp4_african_p <- qp4_african_p %>% mutate(pop3 = case_when(
  pop3 == "MerinaB" ~ "Merina B",
  TRUE ~ pop3
))

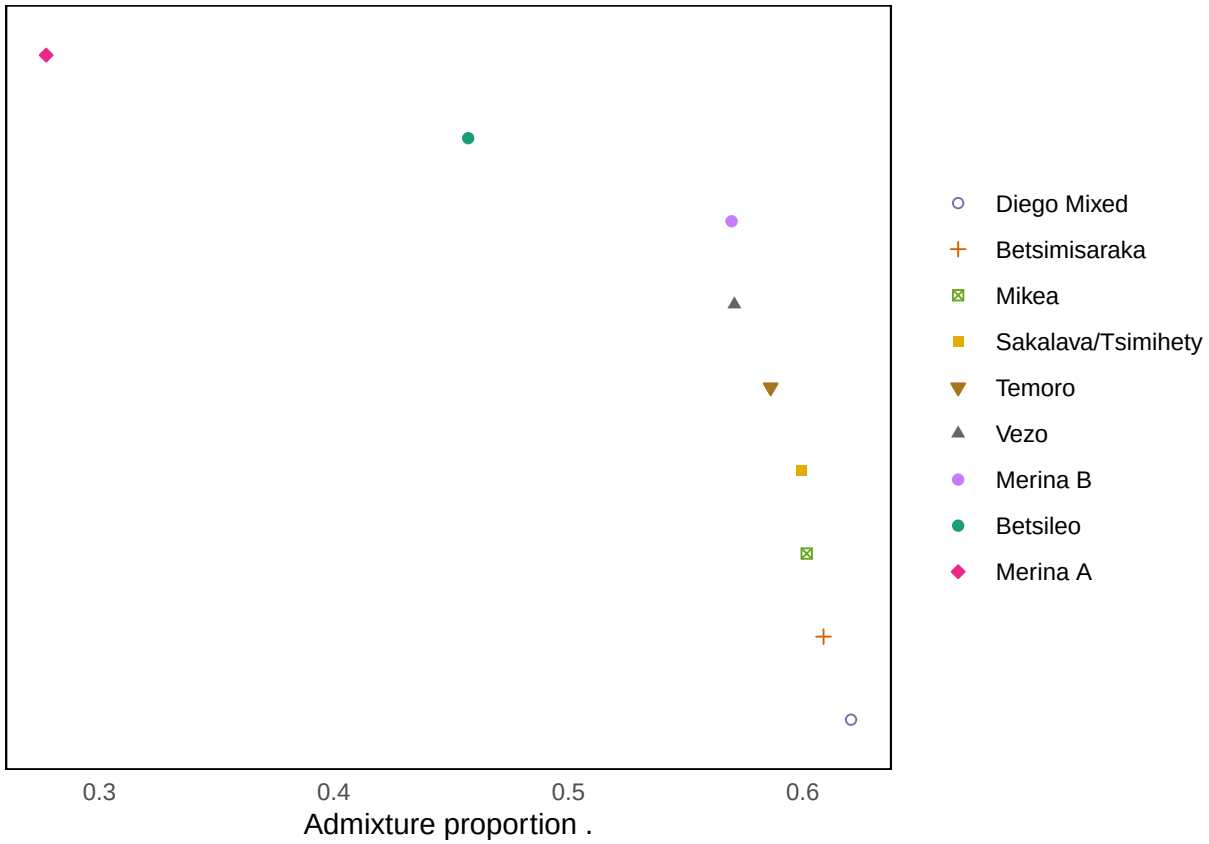
qp4_african_p <- qp4_african_p %>%
  arrange(desc(alpha)) %>%
  mutate(pop3 = forcats::fct_inorder(pop3))
ggplot(qp4_african_p, aes(x = alpha, y = pop3, colour = pop3, fill = pop3)) +
  geom_point() +
  labs(
    x = paste("Admixture proportion \u03b1"), y = NULL,
    colour = "Malagasy population", fill = "Malagasy population"
  ) +
  guides(colour = "none", fill = "none") +
  theme_minimal() +
  theme(
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```



```
ggsave(
  filename = paste0(res_dir, "F4_African_source.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```

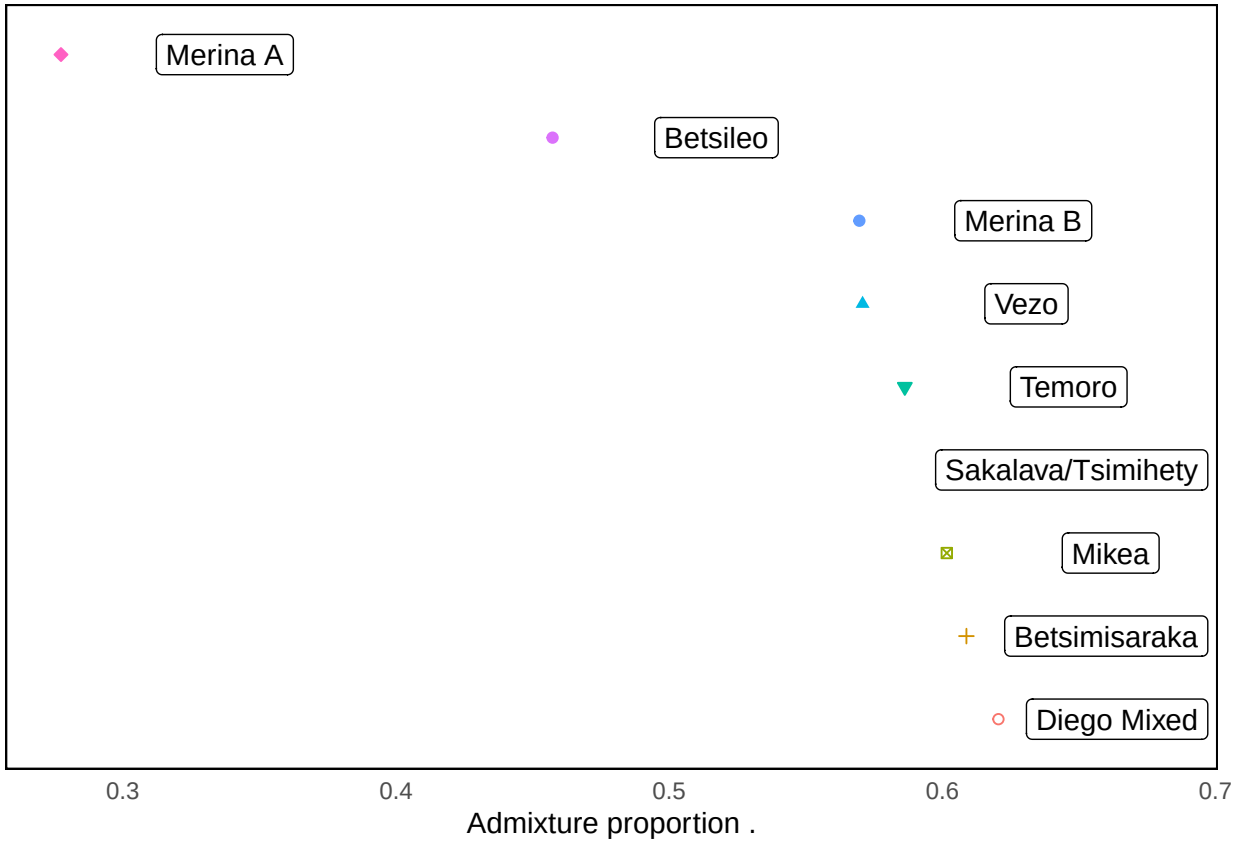
```
ggplot(
  qp4_african_p,
  aes(
    x = alpha, y = pop3, colour = pop3,
    fill = pop3, shape = pop3
  )
) +
  geom_point() +
  labs(
    x = paste("Admixture proportion \u03b1"), y = NULL,
    colour = NULL, fill = NULL, shape = NULL
  ) +
  guides() +
  theme_minimal() +
  theme(
    legend.position.inside = c(0.15, 0.35),
    axis.text.y = element_blank(),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  ) +
  scale_colour_manual(
    values = c(
      "Betsileo" = "#1B9E77",
      "Betsimisaraka" = "#D95F02",
      "Diego Mixed" = "#7570B3",
      "Merina A" = "#E7298A",
      "Merina B" = "#C77CFF",
      "Mikea" = "#66A61E",
      "Sakalava/Tsimihety" = "#E6AB02",
      "Temoro" = "#A6761D",
      "Vezo" = "#666666"
    )
  ) +
  scale_fill_manual(
    values = c(
      "Betsileo" = "#1B9E77",
      "Betsimisaraka" = "#D95F02",
      "Diego Mixed" = "#7570B3",
      "Merina A" = "#E7298A",
      "Merina B" = "#C77CFF",
      "Mikea" = "#66A61E",
      "Sakalava/Tsimihety" = "#E6AB02",
      "Temoro" = "#A6761D",
      "Vezo" = "#666666"
    )
  ) +
```

```
  scale_shape_manual(values = c(1, 3, 7, 15, 25, 17, 19, 21, 23))
```




```
ggplot(
  qp4_african_p,
  aes(
    x = alpha, y = pop3, colour = pop3,
    fill = pop3, shape = pop3
  )
) +
  geom_point() +
  labs(
    x = paste("Admixture proportion \u03b1"), y = NULL,
    colour = NULL, fill = NULL, shape = NULL
  ) +
  guides() +
  theme_minimal() +
  theme(
    legend.position = "none",
    axis.text.y = element_blank(),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  ) +
  scale_shape_manual(values = c(1, 3, 7, 15, 25, 17, 19, 21, 23)) +
```

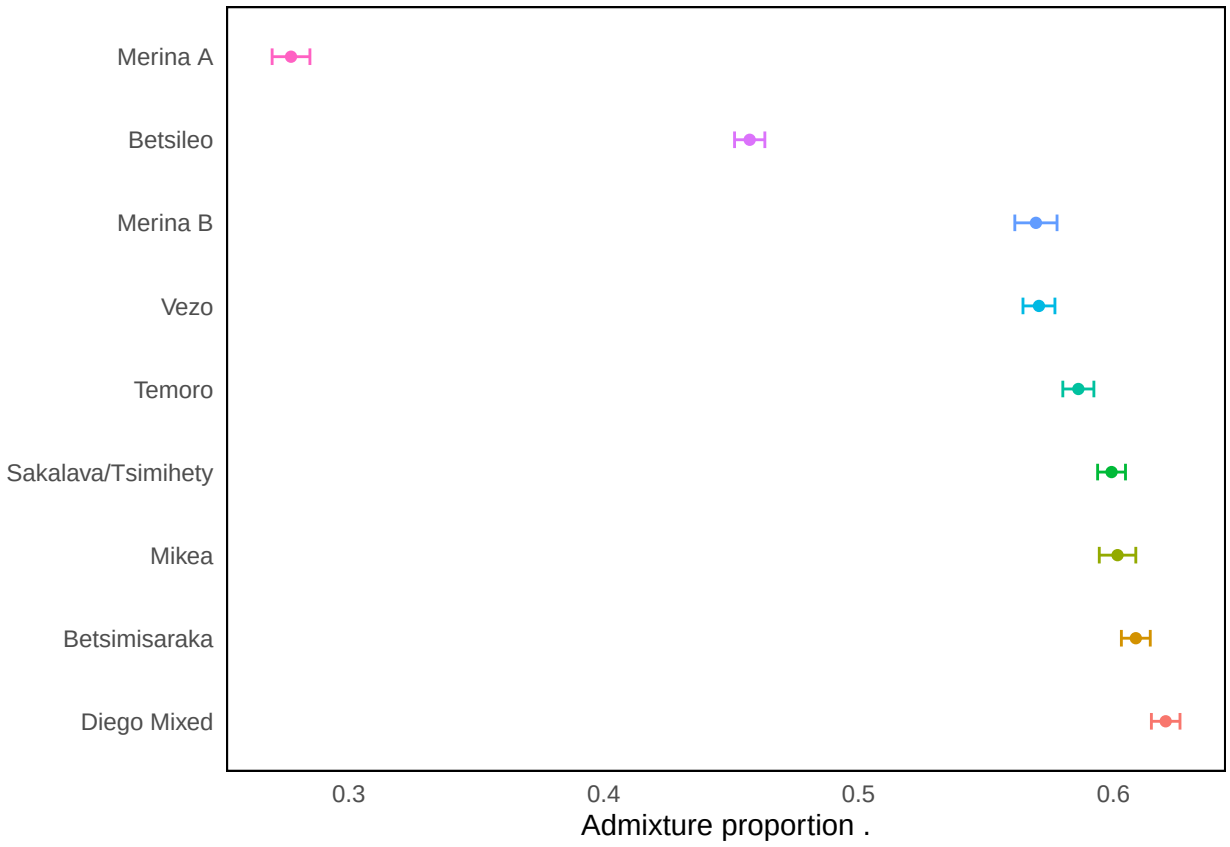
```
  geom_label_repel(
    data = qp4_african_p,
    aes(
      x = alpha, y = pop3, label = pop3,
      fill = NULL, colour = NULL, shape = NULL
    ),
    segment.color = "transparent", nudge_x = 0.06,
    nudge_y = 0, show.legend = FALSE
  )
```




```
qp4_african_p$lower <- qp4_african_p$alpha - qp4_african_p$se
qp4_african_p$upper <- qp4_african_p$alpha + qp4_african_p$se


ggplot(qp4_african_p, aes(x = alpha, y = pop3, colour = pop3, fill = pop3)) +
  geom_point() +
  labs(
    x = paste("Admixture proportion \u03b1"), y = NULL,
    colour = "Malagasy population", fill = "Malagasy population"
  ) +
  guides(colour = "none", fill = "none") +
  theme_minimal() +
  theme(
    legend.position = "right",
    axis.line = element_blank(),
```

```
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  ) +
  geom_errorbar(aes(xmin = lower, xmax = upper),
    width = 0.2
  )
```




```
ggsave(
  filename = paste0(res_dir, "F4_African_source_error_bar.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)


qp4_african_p$region <- c(rep("coastal", 6), rep("highland", 3))
```

Merina is lowest, as expected.

```
pop_matrix <-
  as.matrix(data.frame("GBR", "chimp_ref", malagasy, "ESN", "Borneo"))

qp4_asian_p <- qpf4ratio(f2_blocks, pop_matrix)

qp4_asian_p$pop3 <- factor(
  qp4_asian_p$pop3,
```

```
  c(
    "Betsimisaraka", "MerinaA", "MerinaB", "Diego",
    "Sakalava_Tsimihety", "Betsileo", "Temoro",
    "Vezo", "Mikea"
  )
)

qp4_asian_p <- qp4_asian_p %>% mutate(pop3 = forcats::fct_inorder(pop3))
qp4_asian_p
```

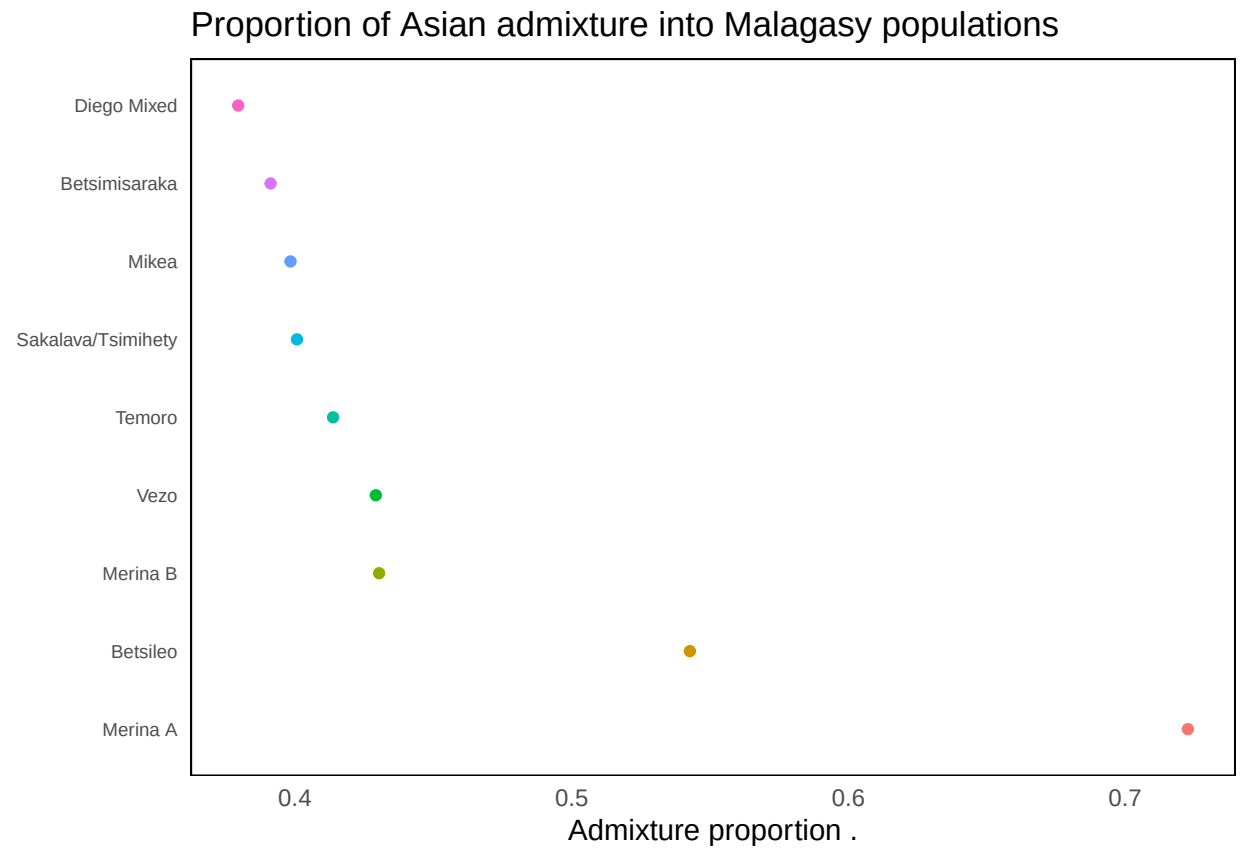
```
## # A tibble: 9 x 8
##   pop1  pop2      pop3               pop4  pop5   alpha      se     z
##   <chr> <chr>     <fct>              <chr> <chr>  <dbl>   <dbl> <dbl>
## 1 GBR   chimp_ref Betsimisaraka      ESN   Borneo 0.391 0.00567  69.0
## 2 GBR   chimp_ref MerinaA            ESN   Borneo 0.723 0.00741  97.5
## 3 GBR   chimp_ref MerinaB            ESN   Borneo 0.430 0.00828  52.0
## 4 GBR   chimp_ref Diego              ESN   Borneo 0.379 0.00561  67.6
## 5 GBR   chimp_ref Sakalava_Tsimihety ESN   Borneo 0.401 0.00547  73.3
## 6 GBR   chimp_ref Betsileo           ESN   Borneo 0.543 0.00594  91.3
## 7 GBR   chimp_ref Temoro             ESN   Borneo 0.414 0.00610  67.9
## 8 GBR   chimp_ref Vezo               ESN   Borneo 0.429 0.00627  68.5
## 9 GBR   chimp_ref Mikea              ESN   Borneo 0.398 0.00717  55.6
```

Merina and Betsileo are highest, as expected.

```
# update labels
# rename qp4_asian_p$pop3 "Sakalava Tsimihety" to "Sakalava/Tsimihety"
qp4_asian_p <- qp4_asian_p %>% mutate(pop3 = case_when(
  pop3 == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
  TRUE ~ pop3
))
# rename qp4_asian_p$pop3 "Diego" to "Diego Mixed"
qp4_asian_p <- qp4_asian_p %>% mutate(pop3 = case_when(
  pop3 == "Diego" ~ "Diego Mixed",
  TRUE ~ pop3
))
# rename qp4_asian_p$pop3 "MerinaA" to "Merina A"
qp4_asian_p <- qp4_asian_p %>% mutate(pop3 = case_when(
  pop3 == "MerinaA" ~ "Merina A",
  TRUE ~ pop3
))
# rename qp4_asian_p$pop3 "MerinaB" to "Merina B"
qp4_asian_p <- qp4_asian_p %>% mutate(pop3 = case_when(
  pop3 == "MerinaB" ~ "Merina B",
  TRUE ~ pop3
))

qp4_asian_p <- qp4_asian_p %>%
  arrange(desc(alpha)) %>%
  mutate(pop3 = forcats::fct_inorder(pop3))
ggplot(qp4_asian_p, aes(x = alpha, y = pop3, colour = pop3, fill = pop3)) +
  geom_point() +
  labs(
```

```
    title = "Proportion of Asian admixture into Malagasy populations",
    x = paste("Admixture proportion \u03b1"), y = NULL,
    colour = "Malagasy population", fill = "Malagasy population"
  ) +
  guides(colour = "none", fill = "none") +
  theme_minimal() +
  theme(
    axis.text.y = element_text(size = 7),
    legend.position = "right",
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```




```
ggsave(
  filename = paste0(res_dir, "F4_Asian_source.jpg"),
  width = 7, height = 5, units = "in", dpi = 300
)
```